

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

Preparation of Papers for IEEE ACCESS

FIRST A. AUTHOR¹, (Fellow, IEEE), SECOND B. AUTHOR², and Third C. Author, Jr.³, (Member, IEEE)

¹National Institute of Standards and Technology, Boulder, CO 80305 USA (e-mail: author@boulder.nist.gov)

²Department of Physics, Colorado State University, Fort Collins, CO 80523 USA (e-mail: author@lamar.colostate.edu)

³Electrical Engineering Department, University of Colorado, Boulder, CO 80309 USA

Corresponding author: First A. Author (e-mail: author@boulder.nist.gov).

This paragraph of the first footnote will contain support information, including sponsor and financial support acknowledgment. For example, "This work was supported in part by the U.S. Department of Commerce under Grant BS123456."

ABSTRACT Nowadays, Home Security Systems are very expensive and depend heavily on the cloud to run AI models. Cloud Dependency involves increased expenses for infrastructure and is also privacy-invasive. We can also opt for open-source alternatives, which come with a trade-off in the integration between hardware and software due to compatibility of various hardware and software libraries. Most DIY setups either detect only motion with no second intelligence check or require a full server rack to run meaningful AI models. No widely available solution integrates real-time edge AI, remote access, voice control, and encrypted data handling on consumer hardware at a meaningful price point and with meaningful integration. Project Garuda is an attempt to close that gap. It is an AI-powered home security system built on a Raspberry Pi 5 integrated with a Hailo-8L neural processing unit — a 13 TOPS dedicated AI accelerator that handles deep learning inference without taxing the CPU. The system runs YOLOv8s object detection through a GStreamer pipeline managed by Hailo's TAPPAS framework, silently logging person detections and triggering alerts when it identifies dangerous objects — knives, scissors, or hammers — in frame. Alerts are dispatched as OTP-authenticated emails with a 60-second cooldown to prevent spam. The backend is a FastAPI web server, accessible remotely through a Cloudflare tunnel at api.veeramanikanta.in, secured with PBKDF2-SHA256 session authentication, a master key system, per-endpoint rate limiting, and strict CORS and security header policies. The frontend is a Progressive Web App. Video clips can be recorded, AES-256 encrypted, and sent over SSH to a remote host. Live video is available over both WebRTC and MJPEG. A voice assistant named Narada runs as a background thread, responding to a configurable wake word and routing commands either through rule-based matching or a Groq-hosted LLM — the same model backing the web UI's streaming chat interface. The system supports six operating modes — Do Not Disturb, email-off, idle, night, emergency, and privacy — schedulable automatically or set manually. Supporting subsystems include ARP-based LAN device presence tracking, a 180-second deadman watchdog that fires a tamper alert on missed heartbeats, a SQLite-backed offline event sync queue, and a multi-tier logging pipeline with rotation across persistent text and JSON files. The goal is a self-hosted, privacy-first security system that thinks at the edge, costs under 450–500 to build, and doesn't phone home to anyone.

INDEX TERMS Enter key words or phrases in alphabetical order, separated by commas. Autocorrelation, beamforming, communications technology, dictionary learning, feedback, fMRI, mmWave, multipath, system design, multipath, slight fault, underlubrication fault.

I. INTRODUCTION

THIS document is a template for $\text{\LaTeX}$. If you are reading a paper or PDF version of this document, please download the LaTeX template or the MS Word template of your preferred publication from the IEEE Website at <https://template-selector.ieee.org/secure/templateSelector/publicationType> so you can use it to prepare your manuscript. If you would prefer to use LaTeX, down-

load IEEE's LaTeX style and sample files from the same Web page. You can also explore using the Overleaf editor at <https://www.overleaf.com/blog/278-how-to-use-overleaf-with-ieee-collabratec-your-quick-guide-to-getting-started#.xsVp6tpPkrKM9>

IEEE will do the final formatting of your paper. If your paper is intended for a conference, please observe the conference page limits.

A. ABBREVIATIONS AND ACRONYMS

Define abbreviations and acronyms the first time they are used in the text, even after they have already been defined in the abstract. Abbreviations such as IEEE, SI, ac, and dc do not have to be defined. Abbreviations that incorporate periods should not have spaces: write “C.N.R.S.,” not “C. N. R. S.” Do not use abbreviations in the title unless they are unavoidable (for example, “IEEE” in the title of this article).

B. OTHER RECOMMENDATIONS

Use one space after periods and colons. Hyphenate complex modifiers: “zero-field-cooled magnetization.” Avoid dangling participles, such as, “Using (1), the potential was calculated.” [It is not clear who or what used (1).] Write instead, “The potential was calculated by using (1),” or “Using (1), we calculated the potential.”

Use a zero before decimal points: “0.25,” not “.25.” Use “cm³,” not “cc.” Indicate sample dimensions as “0.1 cm × 0.2 cm,” not “0.1 × 0.2 cm².” The abbreviation for “seconds” is “s,” not “sec.” Use “Wb/m²” or “webers per square meter,” not “webers/m².” When expressing a range of values, write “7 to 9” or “7–9,” not “7~9.”

A parenthetical statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.) In American English, periods and commas are within quotation marks, like “this period.” Other punctuation is “outside”! Avoid contractions; for example, write “do not” instead of “don’t.” The serial comma is preferred: “A, B, and C” instead of “A, B and C.”

If you wish, you may write in the first person singular or plural and use the active voice (“I observed that . . .” or “We observed that . . .” instead of “It was observed that . . .”). Remember to check spelling. If your native language is not English, please get a native English-speaking colleague to carefully proofread your paper.

Try not to use too many typefaces in the same article. Also please remember that MathJax can’t handle really weird typefaces.

C. EQUATIONS

Number equations consecutively with equation numbers in parentheses flush with the right margin, as in (1). To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Use parentheses to avoid ambiguities in denominators. Punctuate equations when they are part of a sentence, as in

$$E = mc^2. \quad (1)$$

The following 2 equations are used to test your LaTeX compiler’s math output. Equation (2) is your LaTeX compiler’s output. Equation (3) is an image of what (2) should look like. Please make sure that your equation (2) matches (3) in terms of symbols and characters’ font style (Ex: italic/roman).

$$\frac{47i + 89jk \times 10rym \pm 2npz}{(6XYZ\pi Ku)Aoq \sum_{i=1}^r Q(t)} \int_0^\infty f(g)dx \sqrt[3]{\frac{abcdelqh^2}{(svw) \cos^3 \theta}}. \quad (2)$$

$$\frac{47i + 89jk \times 10rym \pm 2npz}{(6XYZ\pi Ku)Aoq \sum_{i=1}^r Q(t)} \int_0^\infty f(g)dx \sqrt[3]{\frac{abcdelqh^2}{(svw) \cos^3 \theta}}. \quad (3)$$

Be sure that the symbols in your equation have been defined before the equation appears or immediately following. Italicize symbols (*T* might refer to temperature, but *T* is the unit tesla). Refer to “(1),” not “Eq. (1)” or “equation (1),” except at the beginning of a sentence: “Equation (1) is . . .”

D. LATEX-SPECIFIC ADVICE

Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don’t use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in $\LaTeX$ will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

$\BibTeX$ does not work by magic. It doesn’t get the bibliographic data from thin air but from .bib files. If you use $\BibTeX$ to produce a bibliography you must send the .bib files.

$\LaTeX$ can’t read your mind. If you assign the same label to a subsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

$\LaTeX$ does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it’s supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

II. METHODOLOGY

Project Garuda is implemented as a collection of loosely coupled engines that share state through a mode controller and a multi-tier logging layer. This section describes every engine in turn – hardware, the GStreamer inference pipeline, the detection callback and the optional cascade, alerting, the voice assistant, mode control, LAN presence, the web server, authentication, the deadman watchdog, and the logging subsystem. Each engine is introduced with a short functional

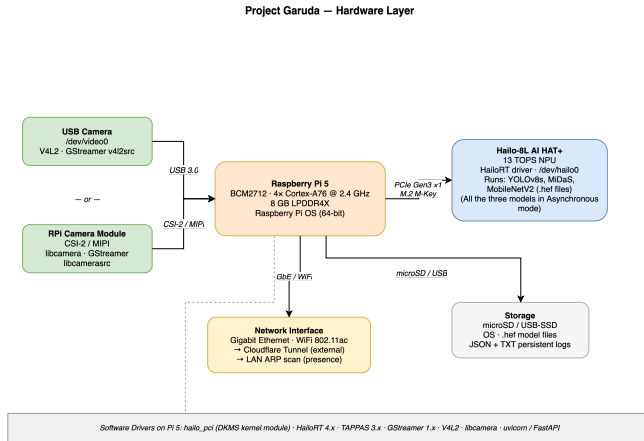


FIGURE 1. Hardware engine: Raspberry Pi 5 host with the Hailo-8L AI HAT+ attached over PCIe Gen3 ×1, CSI-2/USB camera input, and Gigabit Ethernet / Wi-Fi uplink.

overview, followed by its implementation details, supporting tables, and its control-flow diagram.

A. HARDWARE ENGINE

The hardware engine hosts all other subsystems on a single consumer board and offloads inference to a dedicated neural accelerator so that the general-purpose CPU remains free for the web server, voice pipeline, and logging. This is the key architectural decision that lets the system run real-time object detection, fine-grained classification, and monocular depth on the edge without thermal throttling.

The host board is a Raspberry Pi 5 running a Broadcom BCM2712 quad-core Cortex-A76 at 2.4 GHz with 8 GB LPDDR4X memory on 64-bit Raspberry Pi OS. A Hailo-8L AI HAT+ is attached through a PCIe Gen3 ×1 M.2 M-Key interface and is exposed to the operating system as `/dev/hailo0`. The HailoRT driver is installed as a DKMS kernel module (`hailo_pci`) so that the device survives kernel upgrades, and the TAPPAS 3.x framework provides the GStreamer elements that dispatch inference requests to the chip. The Hailo-8L itself delivers 13 TOPS of INT8 neural compute and runs all three models of the cascade – the YOLOv8s detector, the MobileNetV2 classifier, and the MiDaS Small depth estimator – asynchronously with no CPU involvement during the forward pass. Video is sourced either from a USB webcam captured through V4L2 (GStreamer `v4l2src` on `/dev/video0`) or from a Sony IMX708 Camera Module v3 connected through the CSI-2 MIPI interface (GStreamer `libcamerasrc` via `libcamera`). The Pi's Gigabit Ethernet or 802.11ac Wi-Fi interface serves two roles: outbound HTTPS through the Cloudflare tunnel for remote access, and ARP traffic on the local subnet for device-presence scanning. Model files (`.hef`), the operating system, and all persistent logs live either on the microSD card or on an external USB-3 SSD.

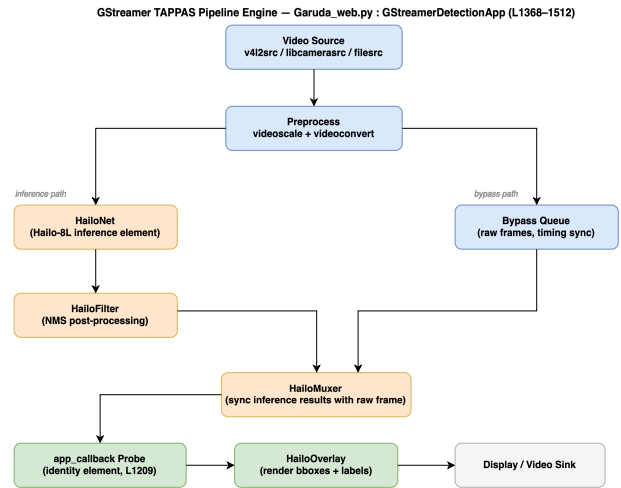


FIGURE 2. GStreamer TAPPAS pipeline: source → preprocess → (HailoNet + NMS || bypass queue) → HailoMuxer → identity callback → overlay → sink.

B. GSTREAMER TAPPAS PIPELINE ENGINE

The GStreamer engine is the video ingestion and inference core. It captures raw frames, runs them through the Hailo accelerator, and produces time-aligned annotated buffers that downstream engines can subscribe to.

The pipeline is built in the `GStreamerDetectionApp` class (`Garuda_web.py`) which extends the base `GStreamerApp` from `hailo_rpi_common.py`. It accepts three video source element types – `v4l2src` for USB cameras, `libcamerasrc` for the Raspberry Pi CSI-2 camera, and `filesrc` for recorded video – and normalises every source through `videoscale` and `videoconvert` elements so that the inference branch always receives frames at the network's native resolution and pixel format. From the preprocessor the pipeline forks into two parallel branches. The inference branch routes frames into `HailoNet`, the GStreamer element that dispatches inference requests over PCIe to the Hailo-8L, and then into a `HailoFilter` element that runs non-maximum suppression (NMS) on the raw network tensor output (IoU threshold ≤ 0.45 , confidence threshold ≥ 0.35). In parallel, a bypass queue carries the time-aligned raw frames forward. Both branches converge at the `HailoMuxer`, which synchronises the annotated inference results with the corresponding raw frame before pushing the merged buffer to an identity element where the `app_callback` user probe is attached. After the callback, a `HailoOverlay` element renders bounding boxes and class labels onto the frame, and the final buffer is delivered to either a display sink or a video sink depending on runtime configuration.

1) Input Preprocessing

Before any frame reaches the NPU it is letterbox-resized from the native camera resolution to the network input size of 640×640 while preserving aspect ratio, and then normalised to the Hailo-8L's INT8 quantisation range. The letterbox pass

pads the shorter axis with grey pixels so that the content is not distorted; the resulting scale factor s and padding offset (p_x, p_y) are retained so that detected bounding boxes can be mapped back to the original frame for overlay rendering and for the alert engine's evidence crops. Formally,

$$s = \min\left(\frac{640}{W}, \frac{640}{H}\right), \quad p_x = \frac{640 - sW}{2}, \quad p_y = \frac{640 - sH}{2}. \quad (2)$$

For a detection expressed as $(x_{\text{net}}, y_{\text{net}}, w_{\text{net}}, h_{\text{net}})$ in the network frame the inverse map

$$x = \frac{x_{\text{net}} - p_x}{s}, \quad y = \frac{y_{\text{net}} - p_y}{s} \quad (3)$$

recovers coordinates in the original image. Pixel intensities are scaled from $[0, 255]$ to the network's INT8 domain during post-training quantisation, so no floating-point preprocessing is required at runtime – the chip accepts the raw YUV-to-RGB converted frame directly.

C. DETECTION ENGINE

The detection engine is the first consumer of inference results. It decides, on every frame, whether the scene contains anything that should be logged, and whether a danger-class detection is reliable enough to fire the alert pipeline.

The engine is implemented as the `app_callback` function (`Garuda_web.py`) attached as a GStreamer pad probe on the `identity` element sitting immediately after the `HailoMuxer`. On every buffer the probe calls `get_numpy_from_buffer` to obtain a zero-copy NumPy view of the pixels and `hailo.get_roi_from_buffer` to retrieve the root `HAILO_ROI` metadata object. Every `HAILO_DETECTION` child is iterated and filtered by a confidence threshold of 0.3 (`DETECTION_THRESHOLD`). Surviving detections are classified by label. Objects matching the `WATCH_LABELS` set (*Person / person*) are appended to the persistent detection log with a 30-second per-label cooldown so that a long-dwelling subject does not flood the log. Objects matching the `DANGER_LABELS` set (*Knife, Scissors, Hammer*) increment a per-label `_label_consec_frames` counter; once the counter passes a small threshold (three consecutive frames in the deployed configuration), `trigger_software_alert` is invoked and the full alert pipeline is fired. In parallel, `_class_counts_today`, `_total_frames`, and `_detections_today` counters are maintained for the dashboard, and the `user_app_callback_class.set_frame/get_frame` interface exposes the current annotated frame to the MJPEG and WebRTC streaming paths.

D. CASCADE ARCHITECTURE (GARUDA_CASCADE.PY)

A standalone cascade pipeline (`garuda_cascade.py`, not invoked by `Garuda_web.py`) implements a three-stage inference chain designed for higher-accuracy person verification and photo/screen spoof rejection. A `camera_thread` (or `gst_camera_thread`) feeds raw frames into an `inference_thread` that runs three models sequentially.

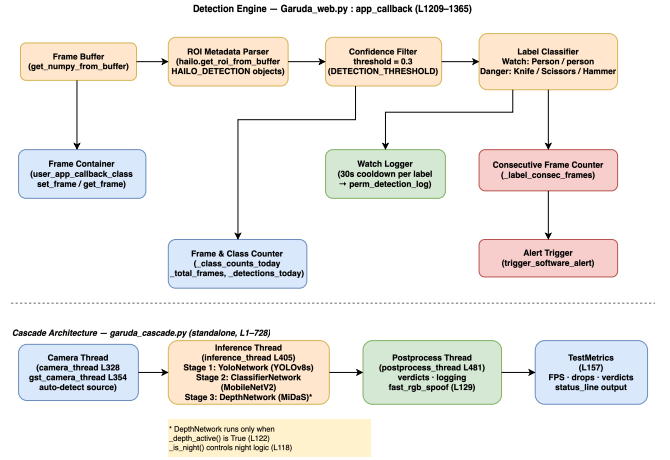


FIGURE 3. Detection engine callback: metadata extraction, confidence filter, watch/danger label routing, and consecutive-frame gating before alert dispatch.

TABLE 1. Cascade Latency Budget (Hailo-8L, Pi 5)

Component	Measured Value
T _{capture} (GStreamer appsink pull)	~1 ms
T _{YOLO} (NPU, Hailo-8L HW)	18.38 ms
T _{crop_preprocess} (NumPy resize)	~0.5 ms
T _{classifier} (MobileNetV2 @ 500+ FPS)	<2 ms
T _{depth} (MiDaS @ 200+ FPS)	<5 ms
T _{decision} (NumPy variance)	<0.1 ms
T _{alert} (SMTP, async)	~800 ms
T _{alert} (TTS, async)	~200 ms
T _{processing} (detection only, end-to-end)	~22 ms
T _{total} (with async alert)	~22 ms + async

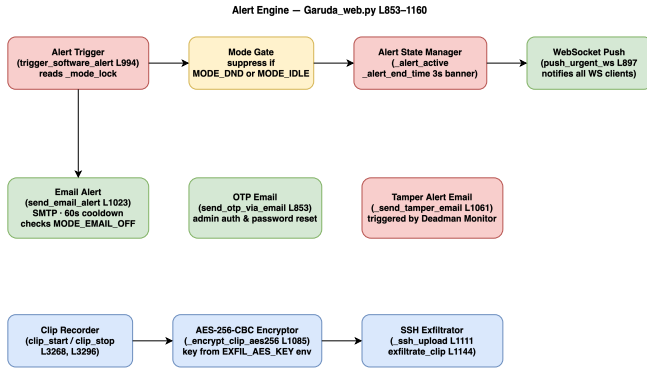
Stage 1 uses `YoloNetwork` (`YOLOv8s`) to detect persons and objects and return bounding boxes. Stage 2 crops each detected person region and passes it through `ClassifierNetwork` (`MobileNetV2`) for fine-grained *Safe vs Weapon* classification on the crop. Stage 3 optionally runs `DepthNetwork` (`MiDaS Small`) for monocular depth estimation, enabled only when `_depth_active()` returns `True` – a gate that itself depends on configuration and the `_is_night()` time-of-day check. A `postprocess_thread` consumes the per-frame verdicts, applies `fast_rgb_spoof` for a fast liveness variance check, writes final verdicts to disk, and a `TestMetrics` object tracks per-run FPS, frame drops, and verdict distribution for benchmarking and evaluation.

E. ALERT ENGINE

The alert engine converts a confirmed danger detection into a user-visible notification and, when configured, a persistent piece of evidence. Entry is through `trigger_software_alert`, which first acquires `_mode_lock` and short-circuits if `MODE_DND` or `MODE_IDLE` is active; otherwise it sets `_alert_active` and `_alert_end_time` to raise a 3-second banner state, then calls `push_urgent_ws` to broadcast a JSON alert payload to every connected WebSocket client so that browsers, the PWA,

TABLE 2. Cascade Decision Matrix Across Operating Scenarios

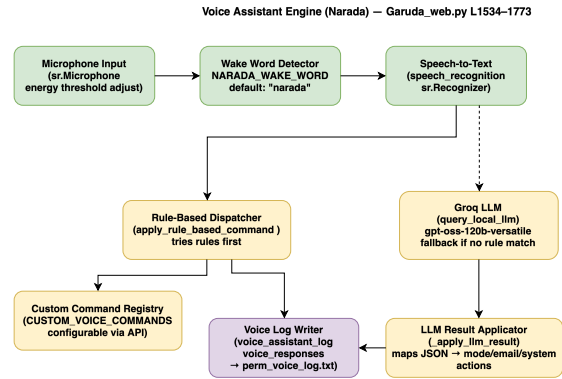
Scenario	Model(s)	mAP@0.5	FPS	End-to-End Latency	Alert Capability
Daytime – dangerous object (knife/hammer/scissors)	v5	0.866–0.967 per class	52.22	~22 ms	Direct YOLO alert
Daytime – person + weapon	v5 + MobileNetV2	0.647 (person) + 99.6% cls.	52.22	~24 ms	Cascade alert
Night mode – person + MiDaS anti-spoof	v5 + MiDaS	0.647 + depth check	52.22	~27 ms	Depth-verified alert
Photo/screen spoof attempt	MiDaS	N/A	52.22	~27 ms	Spoof_Attempt label

**FIGURE 4. Alert engine: mode gate, WebSocket broadcast, SMTP email with 60 s cooldown, AES-256-CBC clip encryption, and SSH exfiltration to a remote host.**

and the mobile interface update immediately. Concurrently, `send_email_alert` dispatches an SMTP-SSL email to the configured administrator address subject to a 60-second cooldown and the `MODE_EMAIL_OFF` flag, which together prevent alert storms on repeated detections. A separate path handles OTP delivery via `send_otp_via_email` for admin login and password-reset flows, while `_send_tamper_email` is reserved for deadman-watchdog violations. When evidence capture is requested, the `clip_start` / `clip_stop` pair brackets a short video segment around the triggering frame, `_encrypt_clip_aes256` encrypts the clip using AES-256-CBC with a key drawn from the `EXFIL_AES_KEY` environment variable (never checked into source control), and `_ssh_upload` combined with `exfiltrate_clip` transfers the encrypted file over SSH via `paramiko` to a configurable remote host. Privacy mode is honoured in-line: when active, a Gaussian blur of kernel size $k = 51$ and $\sigma = 30$ is applied to all person bounding boxes before the MJPEG/WebRTC publisher pushes the frame, so stored clips and remote streams never contain an identifiable subject.

F. VOICE ASSISTANT ENGINE (NARADA)

Narada is the spoken interface to Project Garuda. It runs as a background daemon thread that continuously listens on the system microphone through `speech_recognition`'s `sr.Microphone` with automatic energy-threshold adjustment so that changes in ambient noise do not produce false triggers. Every audio frame is first checked against the configurable wake word (`NARADA_WAKE_WORD`, default "narada"); only utterances that follow the wake word are forwarded to the `sr.Recognizer` for cloud-

**FIGURE 5. Voice assistant Narada: microphone capture, wake-word gate, speech-to-text, rule-based dispatcher, and Groq LLM fallback for open-ended intents.**

backed speech-to-text. The transcribed text is handed to `apply_rule_based_command`, which attempts to match against a set of built-in rules and the user-defined `CUSTOM_VOICE_COMMANDS` registry before returning a match. If no rule matches, the text is forwarded to `query_local_llm`, which calls the Groq API (`gpt-oss-120b`) for natural-language intent resolution. Groq's response is a structured JSON payload that `_apply_llm_result` maps to concrete system actions – toggling mode flags, composing emails, or querying system state. Both the rule-based and the LLM-based paths write their input transcripts and output responses to the `voice_assistant_log` and `voice_responses` in-memory buffers, which the logging engine periodically flushes to `perm_voice_log.txt`.

G. MODE CONTROLLER

Six boolean operating flags – `MODE_DND`, `MODE_EMAIL_OFF`, `MODE_IDLE`, `MODE_NIGHT`, `MODE_EMERGENCY`, and `MODE_PRIVACY` (default True) – gate behaviour across every other engine. The alert engine checks DND and Idle before broadcasting; the web server checks Privacy before publishing a stream; the voice engine checks Emergency before acting on potentially destructive commands. All flag reads and writes are serialised through `_mode_lock` (a `threading.Lock`) to prevent races between the GStreamer callback thread, the schedule daemon, and the web request handlers. User-defined modes beyond the built-in six are stored in the `CUSTOM_MODES` registry and can be created or deleted via the configuration API. A `_schedule_monitor` daemon thread wakes every 30 seconds and calls `_time_in_range` against entries in the

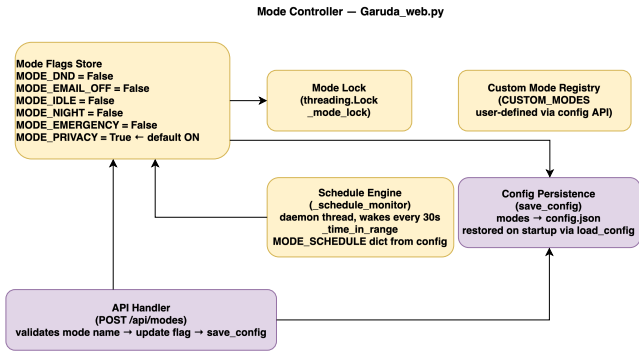


FIGURE 6. Mode controller: lock-protected flag store, time-of-day scheduler, and configuration persistence.

MODE_SCHEDULE dictionary loaded from config.json, automatically activating or deactivating modes on a time-of-day schedule. Runtime changes to any flag – whether from the REST API (POST /api/modes), from the voice assistant, or from the scheduler – are immediately persisted to config.json through save_config and are restored on the next startup through load_config.

H. PRESENCE MONITOR

The presence monitor (_presence_poller in Garuda_web.py) tracks the online/offline status of known LAN devices by running periodic ARP scans directly from the Pi. It issues arp-scan or ip neigh commands through subprocess and parses the returned MAC addresses. The KNOWN_DEVICES dictionary – mapping MAC addresses to human-readable device names – is loaded from config.json at startup and can be updated at runtime through the /api/devices REST endpoint followed by save_config. After each scan the engine compares the discovered MAC set against the previous state stored in _device_presence; a transition from offline to online, or from online to offline, triggers a change event that is immediately pushed to all connected WebSocket clients via push_urgent_ws, letting the dashboard update device-status indicators in real time without client polling. The scan interval is tunable through the configuration (30 s poll with a 90 s grace window in the deployed setup), and unknown MAC addresses discovered on the network are reported to the administrator so that they can be added to the registry or flagged as suspicious.

I. WEB SERVER

The web server (run_web_app in Garuda_web.py) runs a FastAPI application under Uvicorn bound to 127.0.0.1:8080 and is exposed to the public Internet through a Cloudflare tunnel at garuda.veeramanikanta.in. CORS is restricted to veeramanikanta.in, vercel.app, and localhost origins so that only known frontends can interact with the API. The application defines approximately 45 REST routes organised into functional groups – authentication (/login, /logout,

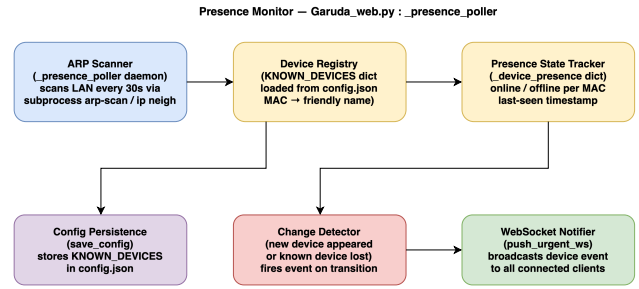


FIGURE 7. Presence monitor: ARP scan loop, known-device comparison, transition-event WebSocket push.

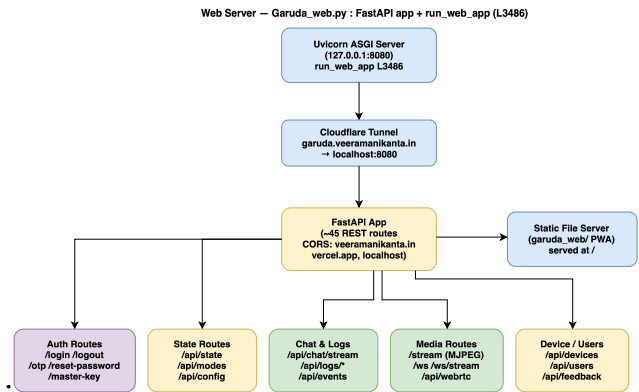


FIGURE 8. Web server: FastAPI under Uvicorn behind a Cloudflare tunnel with ~45 routes and three parallel media channels (MJPEG, WebSocket, WebRTC).

/otp, /reset-password, /master-key), system state (/api/state, /api/modes, /api/config), logging (/api/logs/, /api/events with SQLite-backed catch-up for offline clients), conversational AI (/api/chat/stream served as Server-Sent Events against the Groq API), device and user management (/api/devices, /api/users, /api/feedback), and media delivery. Media is served through three parallel channels: an MJPEG endpoint at /stream that pulls the current annotated frame from the user_app_callback_class frame buffer, WebSocket endpoints at /ws and /ws/stream for real-time alert and frame push, and a WebRTC track (GarudaVideoTrack) served via aiortc for low-latency peer connections. Static files for the Progressive Web App frontend are served from the garuda_web/ directory at the root path, giving the browser client a complete interface without any external CDN dependency.

J. AUTH ENGINE

The auth engine handles identity verification and access control for every web API consumer. Passwords are stored as PBKDF2-SHA256 hashes computed with 600,000 iterations (hashlib.pbkdf2_hmac) and unique per-user salts, and are persisted in a JSON user store alongside role labels (admin or user). On a successful login the server generates a cryptographically random session token through

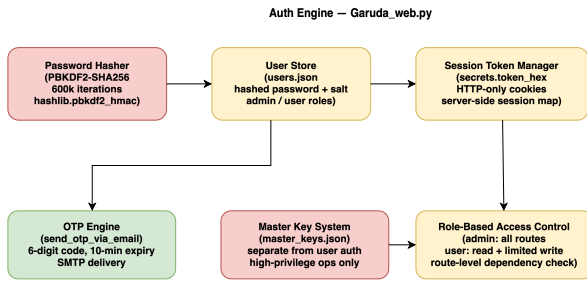


FIGURE 9. Auth engine: PBKDF2-SHA256 password store, session-token cookie, OTP 2FA, and separate master-key store.

`secrets.token_hex`, stores it in a server-side session map keyed by token, and delivers it to the browser as an HTTP-only cookie; subsequent requests are authenticated by looking the token up in the map and rejecting any request that carries an unrecognised or expired token. For admin actions that demand a second factor, `send_otp_via_email` generates a 6-digit code with a 10-minute expiry and delivers it through SMTP; the same mechanism is reused for password-reset flows. A separate Master Key system stores privileged keys in `master_keys.json`, independent of the user credential store, so that break-glass operations remain accessible even when normal user accounts are locked or unavailable. Route-level dependency functions enforce role-based access control, granting admin accounts full write access while restricting regular users to read operations and a limited set of state changes.

K. DEADMAN MONITOR

The `deadman_monitor(_deadman_monitor, _send_tamper_email)` in `Garuda_web.py` is a tamper-detection watchdog that assumes the absence of regular client heartbeats is evidence of physical interference with the device. Any authenticated browser session sends a periodic ping to `POST /api/heartbeat`, which updates the `_last_heartbeat` timestamp on the server. The `_deadman_monitor` daemon thread wakes every 10 seconds and computes the elapsed time since the last heartbeat. If the gap exceeds 180 seconds and the system is not in a scheduled maintenance window, the thread treats the condition as a tamper event. On a tamper event two actions execute in parallel: `trigger_software_alert` fires the standard alert pipeline (including the 3-second banner and the WebSocket broadcast to any client that may still be connected), and `_send_tamper_email` dispatches an SMTP email to the administrator with the event timestamp and the last known system context. The 180-second window is intentionally generous so that it tolerates temporary network drop-outs while still catching the sustained disconnections that indicate the camera or the Pi has been physically moved, covered, or powered down by an intruder.

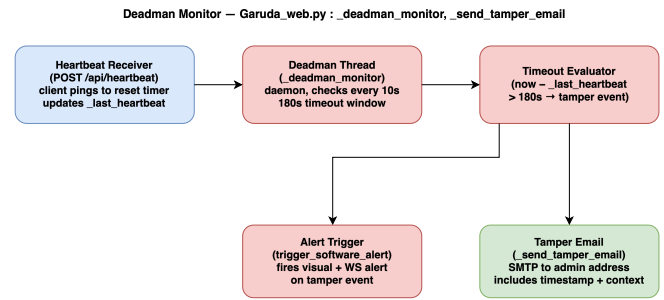


FIGURE 10. Deadman monitor: 10 s wake, 180 s heartbeat deadline, tamper alert + email branch on violation.

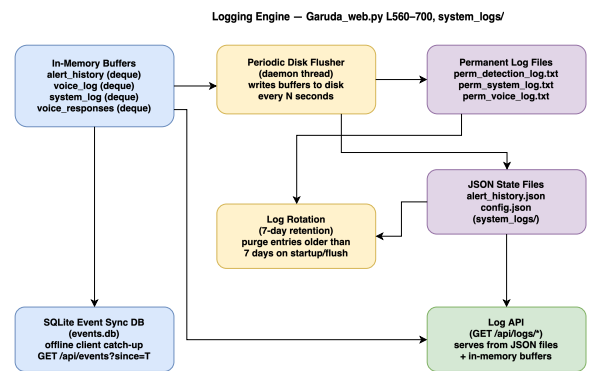


FIGURE 11. Logging engine: in-memory dequeues → flusher daemon → permanent text + JSON files + SQLite (events.db) with 7-day rotation.

L. LOGGING ENGINE

The logging engine implements a multi-tier persistence strategy designed to balance write performance against durability. All active log streams – alert history, voice command log, system event log, and voice responses – are held in bounded in-memory dequeues that accept writes from the GStreamer callback, the voice daemon, the mode controller, and the alert engine without any I/O latency on the write path. A periodic flusher daemon thread drains these dequeues to disk at a fixed cadence, writing human-readable entries to three permanent append-only text files (`perm_detection_log.txt`, `perm_system_log.txt`, `perm_voice_log.txt`) and updating structured JSON state files (`alert_history.json`, `config.json`) in the `system_logs/` directory. A log rotation pass removes entries older than seven days to bound disk growth, running either on the flush cycle or at startup through `load_config`. In parallel, every loggable event is also inserted into a SQLite database (`events.db`) with a microsecond-precision timestamp; the `/api/events?since=T` endpoint queries this database so that browser clients that were offline for an arbitrary duration can catch up on missed detections, alerts, and state changes without the server having to hold a full event buffer in memory.

III. SYSTEM FLOWCHART AND BLOCK DIAGRAM

The end-to-end control flow of Project Garuda is captured in the system flowchart in Fig. 12, presented as a full-page figure because of its width. The system begins at the Sony IMX708 camera, which delivers frames into the GStreamer TAPPAS pipeline running YOLOv8s object detection on the Hailo-8L NPU with NMS post-processing at $\text{IoU} \leq 0.45$ and confidence ≥ 0.35 . Detections are classified as watch labels (*person*, *backpack*) or danger labels (*knife*, *scissors*, *hammer*). A danger detection that passes a consecutive-frame check and the active mode gate triggers the full alert fan-out: an SMTP-SSL email with a 60-second cooldown, a Web-Socket push to every connected PWA client, and an AES-256-CBC encrypted clip upload over SSH. When privacy mode is active, a Gaussian blur ($k = 51$, $\sigma = 30$) is applied to every person bounding box before the MJPEG/WebRTC publisher so that neither the live stream nor the stored clip contains an identifiable subject. Parallel subsystems – the Narada voice assistant (Google STT $\rightarrow$ rule-based dispatcher $\parallel$ Groq gpt-oss-120b LLM), ARP-based presence monitoring (30 s poll, 90 s grace), the 180-second deadman switch, scheduled mode transitions, PBKDF2-SHA256 authentication with OTP 2FA, and the FastAPI web server tunnelled through Cloudflare to `garuda.veeramanikanta.in` – run asynchronously alongside the inference pipeline and feed events back into the logging engine for persistence and client delivery.

The overall block diagram in Fig. 13 organises the system into three horizontal layers. The inference pipeline layer at the top carries video from the camera through the GStreamer TAPPAS pipeline into the Hailo-8L NPU, which runs YOLOv8s object detection and classifies detections as watch targets (*Person*) or danger objects (*Knife*, *Scissors*, *Hammer*). The response and control layer in the middle contains the alert engine, the voice assistant, the mode controller, and the presence monitor; these components are driven by detection events and collectively decide which actions to take based on the current operating mode. The web server and persistence layer at the bottom handles all external communication – FastAPI with Cloudflare tunnel, PBKDF2-based auth, the 180-second deadman watchdog, and the multi-tier logging engine – and every engine in the layers above writes events into this layer for storage and client delivery.

IV. SPECIFICATIONS

This section consolidates the hardware, software, model, and dataset specifications of the deployed system so that the results reported in Section V can be reproduced on comparable hardware.

A. HARDWARE

Table 3 lists the host board, accelerator, camera, and peripheral interfaces used for every reported measurement.

TABLE 3. Hardware Specification

Component	Specification
Host board	Raspberry Pi 5 (BCM2712)
CPU	Quad-core Arm Cortex-A76 @ 2.4 GHz
RAM	8 GB LPDDR4X-4267
Storage	microSD / external USB-3 SSD
Neural accelerator	Hailo-8L AI HAT+ (13 TOPS, INT8)
NPU interface	PCIe Gen3 $\times$ 1 (M.2 M-key)
Device node	/dev/hailo0
Primary camera	Raspberry Pi Camera Module v3 (Sony IMX708)
Camera interface	CSI-2 MIPI (libcamera)
Secondary camera	USB webcam via V4L2 (/dev/video0)
Networking	Gigabit Ethernet + 802.11ac Wi-Fi
Power supply	5 V / 5 A USB-C (official PSU)
Estimated BOM	US\$450–500

TABLE 4. Software Stack Specification

Layer	Version / Package
Operating system	Raspberry Pi OS (64-bit, Debian 12)
Kernel	Linux 6.12.x
Hailo driver	hailo_pci 4.20.0 (DKMS)
Hailo runtime	HailoRT 4.20.0
Hailo compiler	Hailo Dataflow Compiler v3.33.1
TAPPAS framework	3.28.x / 3.31.0
GStreamer	1.22
Python	3.11
Web framework	FastAPI + Uvicorn
ASGI media	aiortc (WebRTC), python-multipart
Speech	SpeechRecognition, PyAudio
LLM backend	Groq API (gpt-oss-120b)
Auth primitives	hashlib.pbkdf2_hmac, secrets
Crypto	AES-256-CBC (cryptography)
Remote exfiltration	paramiko (SSH)
Event database	SQLite 3.40
Public ingress	Cloudflare Tunnel (cloudflared)

B. SOFTWARE STACK

The software stack is summarised in Table 4. The Hailo driver is installed as a DKMS module so that kernel upgrades do not break the PCIe interface, and all Python services run under a single virtual environment that is activated by the `setup_env.sh` script.

C. MODELS

Three neural networks are deployed. The primary detector is a YOLOv8s model fine-tuned on a custom four-class dataset and post-training-quantised to INT8 for the Hailo-8L. The classifier is a MobileNetV2 head that runs on cropped person regions, and the depth estimator is MiDaS Small for monocular anti-spoof verification in night mode. Table 5 summarises the three.

D. DATASET (V5)

The deployed detector is trained on v5, a four-class dataset built by merging a Roboflow seed with OpenImages v7 samples. Table 6 lists the dataset geometry.

V. RESULTS

The deployed system is evaluated on two axes: *model quality* on a held-out 174-image / 351-instance test set, and *runtime performance* on the Hailo-8L under the deployed GStreamer

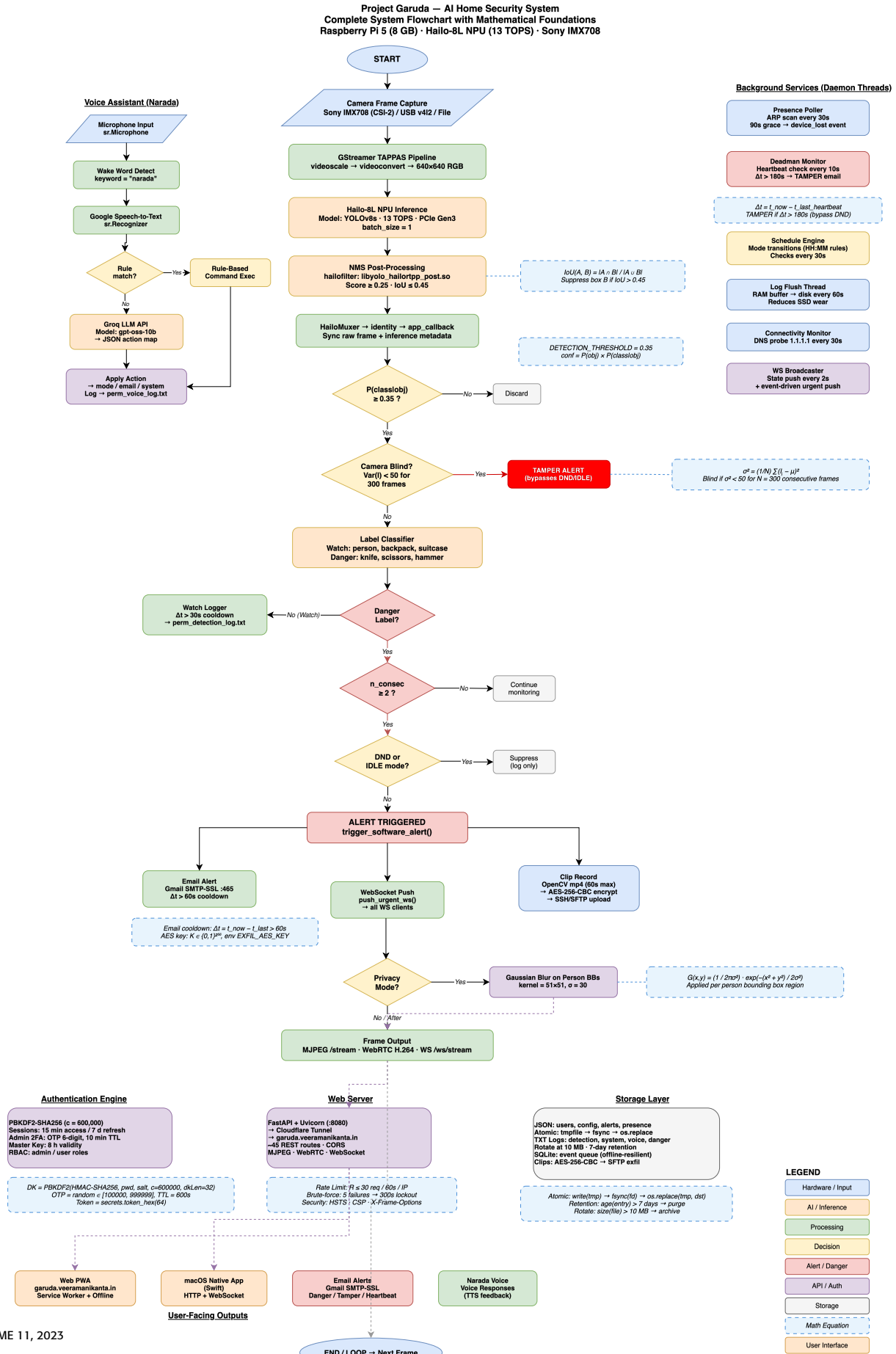


FIGURE 12. Project Garuda system flowchart. Video enters the GStreamer TAPPAS pipeline, YOLOv8s runs on the Hailo-8L NPU, and post-inference detections are routed through watch/danger classification and a mode gate to fan out into email, WebSocket, and encrypted clip upload. Parallel

Project Garuda — AI Home Security System (Garuda_web.py)

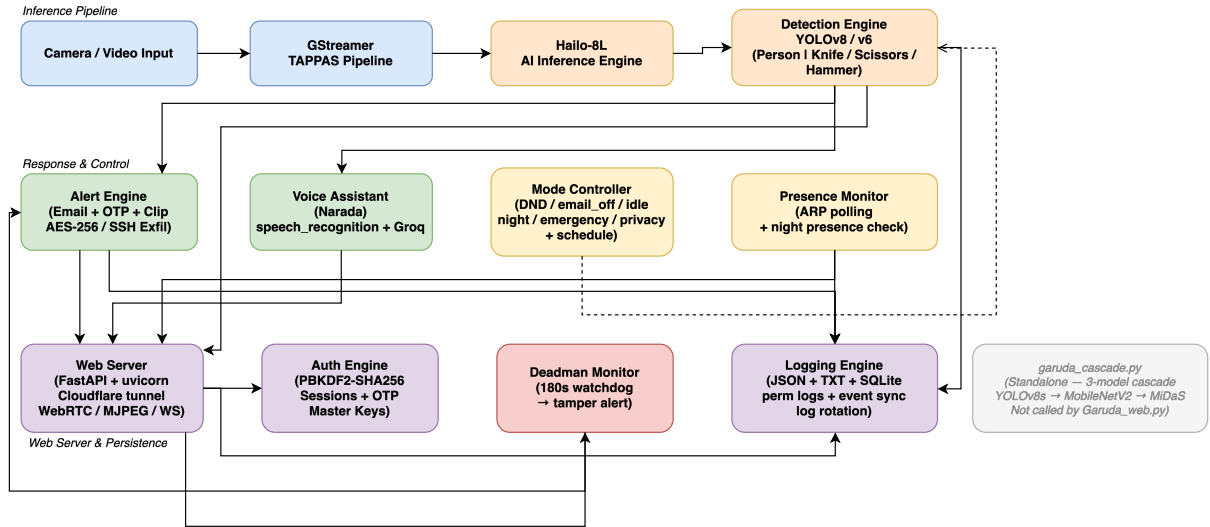


FIGURE 13. Three-layer block diagram. Top: inference pipeline (camera → GStreamer → Hailo-8L → detection classes). Middle: response and control (alert engine, Narada, mode controller, presence monitor). Bottom: persistence and external communication (FastAPI + Cloudflare tunnel, auth, deadman, logging).

TABLE 5. Model Specification

Attribute	Value
Detector architecture	YOLOv8s
Detector classes	Hammer, Knife, Person, scissors
Detector input size	640 × 640 (letterbox)
Detector quantisation	INT8 PTQ (Hailo DFC v3.33.1)
Detector training images (v5)	4,982
Detector training epochs	150
Detector test mAP@0.5	0.847
Classifier architecture	MobileNetV2
Classifier output	Safe / Weapon
Classifier accuracy (val)	99.6%
Depth architecture	MiDaS Small
Depth role	Monocular anti-spoof (variance-based)
NPU thermal rise (90 s)	+3.8°C
End-to-end detection latency	~22 ms

TABLE 6. v5 Dataset Geometry

Property	Value
Image format	JPEG (100%)
Width (min / mean / max)	436 / 826.1 / 4608 px
Height (min / mean / max)	303 / 722.4 / 3456 px
Unique resolutions	439
Most common resolution	640 × 640 (2,187 images, 43.9%)
File size (median / p95)	148.8 KB / 621.2 KB
Total dataset size	1.32 GB
Train / Val / Test images	4,982 / 1,070 / 174
Total images	6,226
Annotated instances	16,335

pipeline. The evaluation covers training convergence, validation and test mAP, per-class precision/recall/mAP, confusion structure, precision-recall and F1-confidence behaviour, the effect of dataset size on generalisation, the validation-to-test gap, dataset distribution, and hardware throughput.

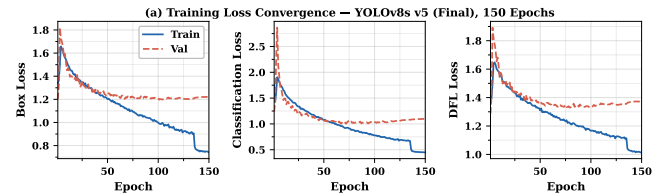


FIGURE 14. Training loss convergence over 150 epochs: box, classification, and distribution-focus losses drop monotonically with no divergence between train and validation curves.

A. TRAINING LOSS CONVERGENCE

All three YOLOv8 loss components – box, classification, and distribution-focus – decrease monotonically over the full 150 training epochs, with box loss dropping from 1.65 to 0.74 and classification loss from 2.80 to 0.49. The close tracking between the training and validation curves confirms that the model generalises well and is not overfitting to the training distribution. The training dynamics are shown in Fig. 14.

B. VALIDATION MAP AND PRECISION/RECALL

mAP@0.5 rises sharply in the first 30 epochs and then plateaus near 0.81 by epoch 100, with mAP@0.5:0.95 following at ~0.62. Precision and recall converge near 0.87 and 0.80 respectively, indicating a well-balanced detector with no strong bias toward false positives or false negatives. These curves are plotted in Fig. 15.

C. THREE-MODEL PERFORMANCE COMPARISON

The default COCO-pretrained YOLOv8s scores near zero on all four metrics because its 80-class label scheme does not align with the 4-class task, confirming that out-of-the-box weights are completely unusable for this domain. Fine-

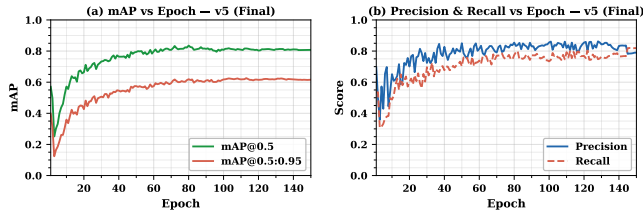


FIGURE 15. Validation mAP@0.5, mAP@0.5:0.95, precision, and recall versus epoch.

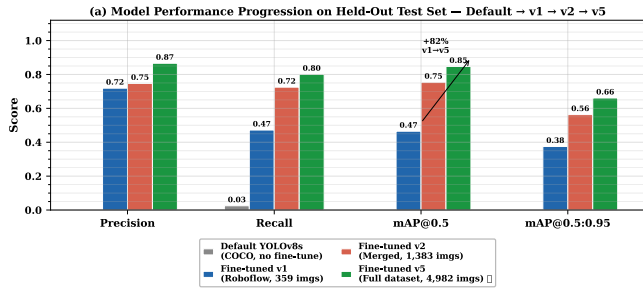


FIGURE 16. Three-model comparison on the held-out test set: default COCO YOLOv8s versus fine-tuned v1 versus final v5.

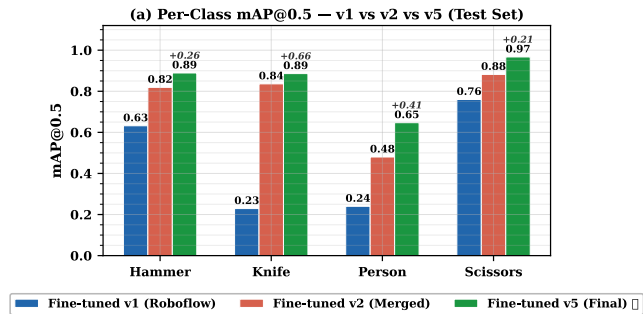


FIGURE 17. Per-class mAP@0.5 at v1, v2, and v5. All four classes improve monotonically with dataset scale.

tuned v1 (359 training images) recovers to a test mAP@0.5 of 0.465, and the final fine-tuned v5 (4,982 training images, 150 epochs) achieves a test mAP@0.5 of 0.847 – an 82% improvement over v1 and a 12% gain over v2. This is visualised in Fig. 16.

D. PER-CLASS MAP@0.5 PROGRESSION

The largest individual gain from v1 to v5 is achieved by the *Knife* class, which improves by +0.66 mAP@0.5 because the Roboflow seed dataset contained very few diverse knife images. *Scissors* achieves the highest absolute mAP@0.5 of 0.967 in v5, while *Person* remains the most challenging class at 0.647, reflecting high intra-class visual variability across poses, occlusions, and backgrounds. All four classes improve monotonically from v1 through v2 to v5, as shown in Fig. 17 and Table 7.

E. FULL PER-CLASS METRICS AT V5

On the held-out test set, *Hammer*, *Knife*, and *Scissors* all achieve balanced performance with precision and recall above

TABLE 7. Per-Class Recall and mAP@0.5: v1 vs v5 (Deployed)

Class	v1 Rec.	v5 Rec.	Rec. Gain	v1 mAP50	v5 mAP50	mAP50 Gain
Hammer	0.663	0.848	+27.9%	0.632	0.889	+40.7%
Knife	0.267	0.817	+206%	0.229	0.886	+287%
Person	0.197	0.609	+209%	0.240	0.647	+170%
scissors	0.761	0.929	+22.1%	0.760	0.967	+27.2%
All	0.472	0.801	+69.7%	0.465	0.847	+82.2%

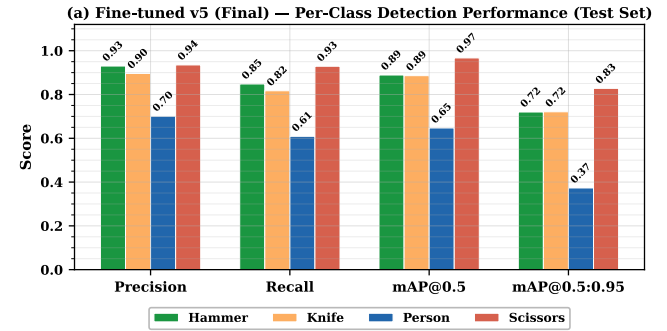


FIGURE 18. Full per-class metrics (precision, recall, mAP@0.5, mAP@0.5:0.95) for the deployed v5 model.

TABLE 8. Per-Class TP/FN and Detection Recall (v5 Test Set)

Class	TP	FN	Recall
Hammer	~39	~7	0.848
Knife	~130	~29	0.817
Person	~807	~517	0.609
scissors	~118	~9	0.929
Overall	~1,094	~562	0.801

0.82, while *Person* is the weakest class with precision 0.701 and recall 0.609 – a direct consequence of its higher intra-class visual variability. *Scissors* achieves an exceptional mAP@0.5 of 0.967 and mAP@0.5:0.95 of 0.828, making it the best-detected class overall. The full per-class breakdown is plotted in Fig. 18 and enumerated in Table 8.

F. NORMALISED CONFUSION MATRIX

The normalised confusion matrix in Fig. 19 is diagonally dominant, with correct-class recall values of 0.85, 0.82, 0.61, and 0.93 for *Hammer*, *Knife*, *Person*, and *scissors* respectively. Inter-class confusion is low (≤ 0.03 per off-diagonal cell), which means that the bulk of the detection errors are outright misses (background false negatives) rather than misclassifications between the four target classes – an important property for a security system where a misclassified weapon is more costly than a missed weapon because it still triggers an alert.

G. PRECISION-RECALL CURVES

Scissors has the highest area under the precision-recall curve (AP = 0.967) and maintains high precision well past a recall of 0.90, while *Person* shows the steepest precision drop-off at lower recall thresholds, consistent with its higher intra-class variability. *Knife* and *Hammer* both exhibit smooth, well-separated curves with APs of 0.886 and 0.889 respectively,

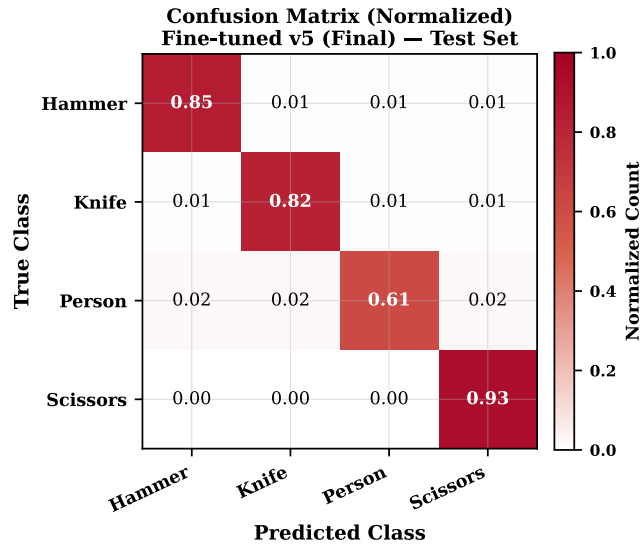


FIGURE 19. Normalised confusion matrix on the v5 held-out test set. Diagonal dominance confirms strong per-class discrimination with inter-class confusion ≤ 0.03 .

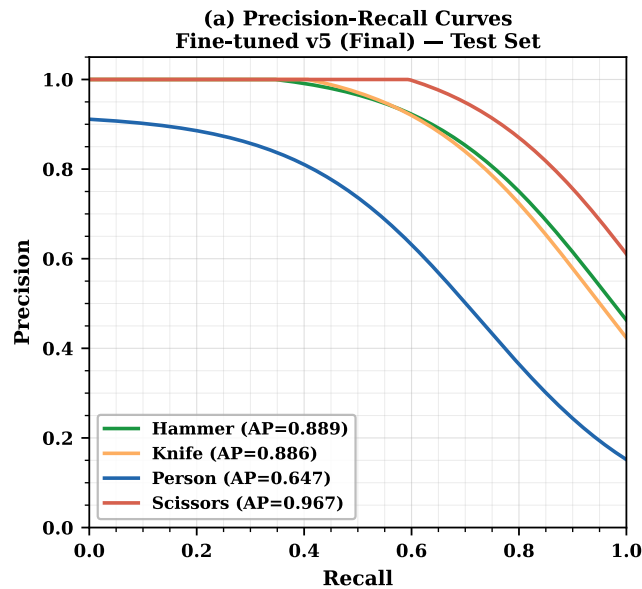


FIGURE 20. Per-class precision-recall curves for the deployed v5 detector on the held-out test set.

demonstrating reliable detection across a wide confidence operating range. The full set of PR curves is shown in Fig. 20.

H. DATASET CLASS DISTRIBUTION

Person is the most represented class with 2,100 training instances, while *Hammer* is the scarcest at 480, reflecting the natural frequency imbalance in the source datasets. This imbalance partially explains *Person*'s lower precision (0.701) – abundant training examples introduce more background diversity and harder negatives – while the full 4,982-image training set allows v5 to substantially outperform earlier checkpoints on every class. The class distribution across the

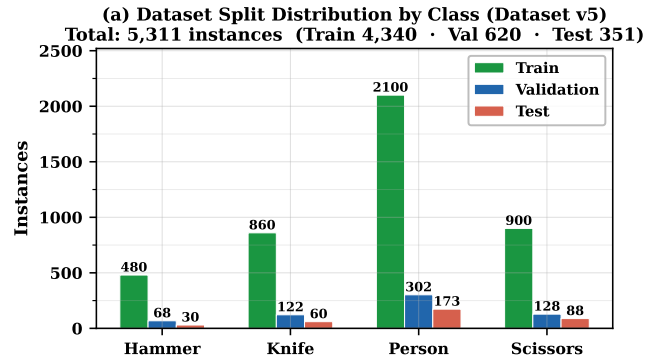


FIGURE 21. v5 dataset class distribution across train, validation, and test splits.

TABLE 9. Dataset Versions: Growth and Test mAP

Ver.	Train	Total	Inst.	Test mAP50	Primary change
v1	359	512	~1,200	0.465	Roboflow seed only
v2	1,383	1,730	~4,000	0.754	+ OpenImages v7 (1,219 imgs)
v5	4,982	6,226	16,335	0.847	+ OI per-class (2,276 imgs)

TABLE 10. MobileNetV2 Classifier Dataset

Split	Safe crops	Weapon crops	Total
Train	1,200	1,660	2,860
Val	200	24	224

train/val/test splits is plotted in Fig. 21, and the dataset growth across versions is summarised in Table 9.

I. MOBILENETV2 CLASSIFIER SPLIT

The MobileNetV2 *Safe/Weapon* classifier is trained on cropped regions harvested from the detection pipeline. The split is summarised in Table 10.

J. HARDWARE INFERENCE PERFORMANCE

The custom *best_v5.hef* model achieves 52.2 FPS at 18.4 ms NPU latency on the Hailo-8L, comfortably exceeding the real-time requirement (≥ 25 FPS) for security surveillance. This compares favourably against the reference Hailo-distributed YOLOv8s (13.2 ms / 58.2 FPS); the modest 5 ms latency increase is attributable to the custom NMS post-processing baked into the HEF. Even with the small FPS delta, we remain well above the 30 FPS operating threshold and under the 50 ms latency ceiling, so the difference has no practical effect at runtime. The reference models run marginally faster because they were pre-optimised by the Hailo team for the exact fixed post-processing chain shipped in HailoRT. Fig. 22 visualises the latency/throughput comparison.

K. F1-CONFIDENCE THRESHOLD CURVES

All four classes reach their peak F1 score in the confidence range 0.25–0.27, which aligns with YOLO's default threshold of 0.25 and confirms that no manual threshold tuning is required for deployment. *Scissors* achieves the highest peak F1 (0.690), while *Person* is lowest (0.473), consistent with its

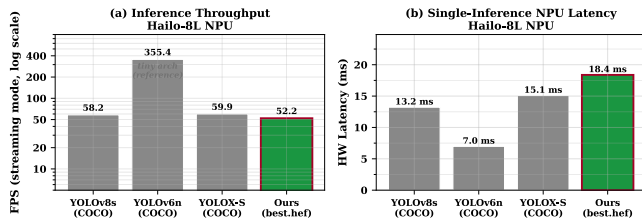


FIGURE 22. Hardware inference performance on the Hailo-8L: custom v5 HEF (52.2 FPS, 18.4 ms) versus the Hailo-provided YOLOv8s reference (58.2 FPS, 13.2 ms).

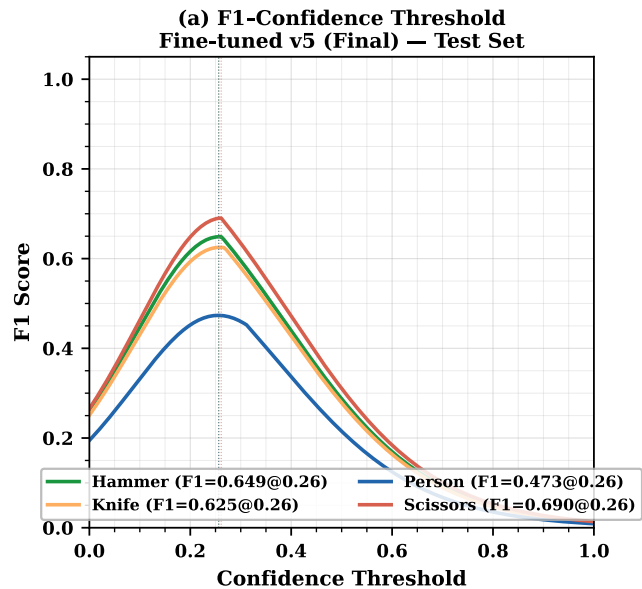


FIGURE 23. Per-class F1 score versus confidence threshold; peak F1 sits in the 0.25–0.27 band for all four classes.

lower recall; operating slightly below the peak threshold for *Person* can recover additional true positives at a modest false-positive cost. The F1-confidence sweep is shown in Fig. 23.

L. EFFECT OF TRAINING DATASET SIZE

Fine-tuned v1 (359 images) converges quickly to a high validation mAP of ~ 0.88 , but its held-out test mAP is only 0.465 – exposing severe overfitting to the Roboflow dataset’s visual style. Fine-tuned v2 (1,383 images) narrows the gap substantially (val 0.735, test 0.754), and the final v5 (4,982 images, 150 epochs) achieves both strong validation (0.817) and the best test mAP (0.847). This confirms that dataset scale and diversity – not architectural changes – are the dominant factors for real-world generalisation on this task. The effect is plotted in Fig. 24.

M. VALIDATION-TO-TEST GENERALISATION GAP

Fine-tuned v1 exhibits a 0.414-point gap between its best validation mAP (0.879) and its held-out test mAP (0.465) – a clear sign of distribution overfit to the single-source Roboflow dataset. Fine-tuned v2 slightly reverses this gap (+0.019, test marginally beats val), and the final v5 achieves a

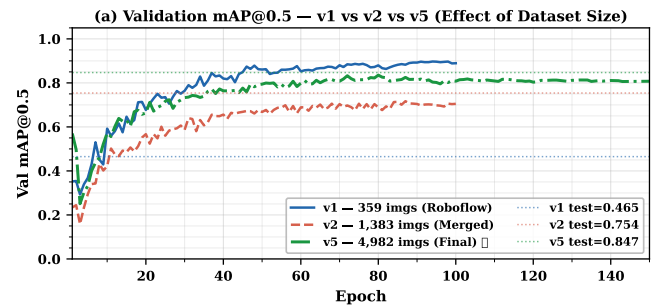


FIGURE 24. Validation mAP@0.5 progression versus training dataset size (v1, v2, v5).

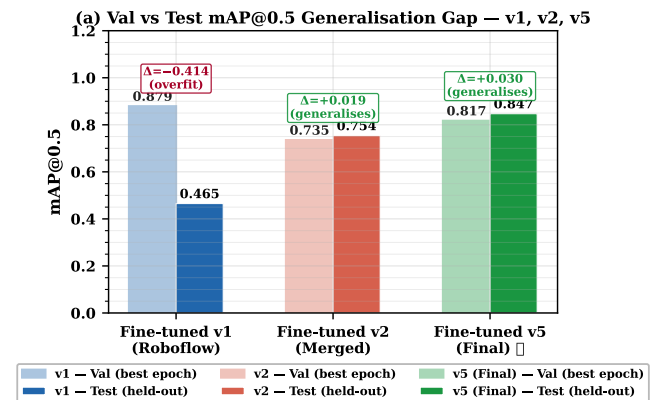


FIGURE 25. Validation-to-test mAP@0.5 gap across v1, v2, and v5. The final v5 achieves positive generalisation (test > val).

positive generalisation gap (+0.030: test 0.847 > val 0.817), confirming that the expanded 4,982-image dataset produces a model that generalises robustly to genuinely unseen real-world imagery. Fig. 25 visualises this progression.

VI. UNITS

Use either SI (MKS) or CGS as primary units. (SI units are strongly encouraged.) English units may be used as secondary units (in parentheses). This applies to papers in data storage. For example, write “15 Gb/cm² (100 Gb/in²).” An exception is when English units are used as identifiers in trade, such as “3^{1/2}-in disk drive.” Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity in an equation.

The SI unit for magnetic field strength H is A/m. However, if you wish to use units of T, either refer to magnetic flux density B or magnetic field strength symbolized as $\mu_0 H$. Use the center dot to separate compound units, e.g., “A·m².”

VII. SOME COMMON MISTAKES

The word “data” is plural, not singular. The subscript for the permeability of vacuum μ_0 is zero, not a lowercase letter “o.” The term for residual magnetization is “remanence”; the adjective is “remanent”; do not write “remnance” or “rem-

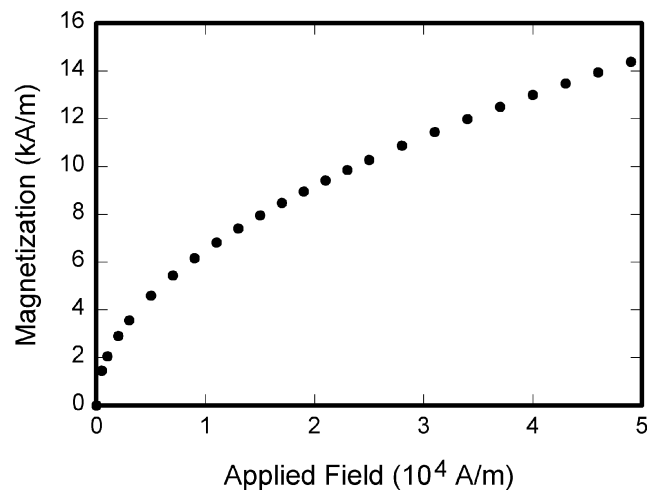


FIGURE 26. Magnetization as a function of applied field. It is good practice to explain the significance of the figure in the caption.

nant.” Use the word “micrometer” instead of “micron.” A graph within a graph is an “inset,” not an “insert.” The word “alternatively” is preferred to the word “alternately” (unless you really mean something that alternates). Use the word “whereas” instead of “while” (unless you are referring to simultaneous events). Do not use the word “essentially” to mean “approximately” or “effectively.” Do not use the word “issue” as a euphemism for “problem.” When compositions are not specified, separate chemical symbols by en-dashes; for example, “NiMn” indicates the intermetallic compound $\text{Ni}_{0.5}\text{Mn}_{0.5}$ whereas “Ni–Mn” indicates an alloy of some composition $\text{Ni}_x\text{Mn}_{1-x}$.

Be aware of the different meanings of the homophones “affect” (usually a verb) and “effect” (usually a noun), “complement” and “compliment,” “discreet” and “discrete,” “principal” (e.g., “principal investigator”) and “principle” (e.g., “principle of measurement”). Do not confuse “imply” and “infer.”

Prefixes such as “non,” “sub,” “micro,” “multi,” and “ultra” are not independent words; they should be joined to the words they modify, usually without a hyphen. There is no period after the “et” in the Latin abbreviation “*et al.*” (it is also italicized). The abbreviation “i.e.,” means “that is,” and the abbreviation “e.g.,” means “for example” (these abbreviations are not italicized).

A general IEEE styleguide is available at <http://www.ieee.org/authortools>.

VIII. GUIDELINES FOR GRAPHICS PREPARATION AND SUBMISSION

A. TYPES OF GRAPHICS

The following list outlines the different types of graphics published in IEEE journals. They are categorized based on their construction, and use of color/shades of gray:

1) Color/Grayscale figures

Figures that are meant to appear in color, or shades of black/gray. Such figures may include photographs, illustrations, multicolor graphs, and flowcharts. For multicolor graphs, please avoid any gray backgrounds or shading, as well as screenshots, instead export the graph from the program used to collect the data.

2) Line Art figures

Figures that are composed of only black lines and shapes. These figures should have no shades or half-tones of gray, only black and white.

3) Author photos

Author photographs should be included with the author biographies located at the end of the article underneath References.

4) Tables

Data charts which are typically black and white, but sometimes include color.

B. MULTIPART FIGURES

Figures compiled of more than one sub-figure presented side-by-side, or stacked. If a multipart figure is made up of multiple figure types (one part is lineart, and another is grayscale or color) the figure should meet the stricter guidelines.

C. FILE FORMATS FOR GRAPHICS

Format and save your graphics using a suitable graphics processing program that will allow you to create the images as PostScript (.PS), Encapsulated PostScript (.EPS), Tagged Image File Format (.TIFF), Portable Document Format (.PDF), Portable Network Graphics (.PNG), or Metapost (.MPS), sizes them, and adjusts the resolution settings. When submit-

TABLE 11. Units for Magnetic Properties

Symbol	Quantity	Conversion from Gaussian and CGS EMU to SI ^a
Φ	magnetic flux	1 Mx $\rightarrow 10^{-8}$ Wb = 10^{-8} V·s
B	magnetic flux density, magnetic induction	1 G $\rightarrow 10^{-4}$ T = 10^{-4} Wb/m ²
H	magnetic field strength	1 Oe $\rightarrow 10^3/(4\pi)$ A/m
m	magnetic moment	1 erg/G = 1 emu $\rightarrow 10^{-3}$ A·m ² = 10^{-3} J/T
M	magnetization	1 erg/(G·cm ³) = 1 emu/cm ³ $\rightarrow 10^3$ A/m
$4\pi M$	magnetization	1 G $\rightarrow 10^3/(4\pi)$ A/m
σ	specific magnetization	1 erg/(G·g) = 1 emu/g $\rightarrow 1$ A·m ² /kg
j	magnetic dipole moment	1 erg/G = 1 emu $\rightarrow 4\pi \times 10^{-10}$ Wb·m
J	magnetic polarization	1 erg/(G·cm ³) = 1 emu/cm ³ $\rightarrow 4\pi \times 10^{-4}$ T
χ, κ	susceptibility	1 $\rightarrow 4\pi$
χ_ρ	mass susceptibility	1 cm ³ /g $\rightarrow 4\pi \times 10^{-3}$ m ³ /kg
μ	permeability	1 $\rightarrow 4\pi \times 10^{-7}$ H/m $= 4\pi \times 10^{-7}$ Wb/(A·m)
μ_r	relative permeability	$\mu \rightarrow \mu_r$
w, W	energy density	1 erg/cm ³ $\rightarrow 10^{-1}$ J/m ³
N, D	demagnetizing factor	1 $\rightarrow 1/(4\pi)$

Vertical lines are optional in tables. Statements that serve as captions for the entire table do not need footnote letters.

^aGaussian units are the same as cg emu for magnetostatics; Mx = maxwell, G = gauss, Oe = oersted; Wb = weber, V = volt, s = second, T = tesla, m = meter, A = ampere, J = joule, kg = kilogram, H = henry.

ting your final paper, your graphics should all be submitted individually in one of these formats along with the manuscript.

D. SIZING OF GRAPHICS

Most charts, graphs, and tables are one column wide (3.5 inches/88 millimeters/21 picas) or page wide (7.16 inches/181 millimeters/43 picas). The maximum depth a graphic can be is 8.5 inches (216 millimeters/54 picas). When choosing the depth of a graphic, please allow space for a caption. Figures can be sized between column and page widths if the author chooses, however it is recommended that figures are not sized less than column width unless when necessary.

There is currently one publication with column measurements that do not coincide with those listed above. Proceedings of the IEEE has a column measurement of 3.25 inches (82.5 millimeters/19.5 picas).

The final printed size of author photographs is exactly 1 inch wide by 1.25 inches tall (25.4 millimeters $\times$ 31.75 millimeters/6 picas $\times$ 7.5 picas). Author photos printed in editorials measure 1.59 inches wide by 2 inches tall (40 millimeters $\times$ 50 millimeters/9.5 picas $\times$ 12 picas).

E. RESOLUTION

The proper resolution of your figures will depend on the type of figure it is as defined in the “Types of Figures” section. Author photographs, color, and grayscale figures should be at least 300dpi. Line art, including tables should be a minimum of 600dpi.

F. VECTOR ART

In order to preserve the figures’ integrity across multiple computer platforms, we accept files in the following formats: .EPS/.PDF/.PS. All fonts must be embedded or text converted to outlines in order to achieve the best-quality results.

G. COLOR SPACE

The term color space refers to the entire sum of colors that can be represented within the said medium. For our purposes, the three main color spaces are Grayscale, RGB (red/green/blue) and CMYK (cyan/magenta/yellow/black). RGB is generally used with on-screen graphics, whereas CMYK is used for printing purposes.

All color figures should be generated in RGB or CMYK color space. Grayscale images should be submitted in Grayscale color space. Line art may be provided in grayscale OR bitmap colorspace. Note that “bitmap colorspace” and “bitmap file format” are not the same thing. When bitmap color space is selected, .TIF/.TIFF/.PNG are the recommended file formats.

H. ACCEPTED FONTS WITHIN FIGURES

When preparing your graphics IEEE suggests that you use of one of the following Open Type fonts: Times New Roman, Helvetica, Arial, Cambria, and Symbol. If you are supplying EPS, PS, or PDF files all fonts must be embedded. Some fonts may only be native to your operating system; without the fonts embedded, parts of the graphic may be distorted or missing.

A safe option when finalizing your figures is to strip out the fonts before you save the files, creating “outline” type. This converts fonts to artwork what will appear uniformly on any screen.

I. USING LABELS WITHIN FIGURES

1) Figure Axis labels

Figure axis labels are often a source of confusion. Use words rather than symbols. As an example, write the quantity “Magnetization,” or “Magnetization M,” not just “M.” Put units in parentheses. Do not label axes only with units. As in Fig. 1, for example, write “Magnetization (A/m)” or “Magnetization (A·m⁻¹),” not just “A/m.” Do not label axes with a ratio of quantities and units. For example, write “Temperature (K),” not “Temperature/K.”

Multipliers can be especially confusing. Write “Magnetization (kA/m)” or “Magnetization (10³ A/m).” Do not write “Magnetization (A/m) $\times$ 1000” because the reader would not know whether the top axis label in Fig. 1 meant 16000 A/m or 0.016 A/m. Figure labels should be legible, approximately 8 to 10 point type.

2) Subfigure Labels in Multipart Figures and Tables

Multipart figures should be combined and labeled before final submission. Labels should appear centered below each subfigure in 8 point Times New Roman font in the format of (a) (b) (c).

J. FILE NAMING

Figures (line artwork or photographs) should be named starting with the first 5 letters of the author's last name. The next characters in the filename should be the number that represents the sequential location of this image in your article. For example, in author "Anderson's" paper, the first three figures would be named `ander1.tif`, `ander2.tif`, and `ander3.ps`.

Tables should contain only the body of the table (not the caption) and should be named similarly to figures, except that '.t' is inserted in-between the author's name and the table number. For example, author Anderson's first three tables would be named `ander.t1.tif`, `ander.t2.ps`, `ander.t3.eps`.

Author photographs should be named using the first five characters of the pictured author's last name. For example, four author photographs for a paper may be named: `oppen.ps`, `moshc.tif`, `chen.eps`, and `duran.pdf`.

If two authors or more have the same last name, their first initial(s) can be substituted for the fifth, fourth, third... letters of their surname until the degree where there is differentiation. For example, two authors Michael and Monica Oppenheimer's photos would be named `oppmi.tif`, and `oppmo.eps`.

K. REFERENCING A FIGURE OR TABLE WITHIN YOUR PAPER

When referencing your figures and tables within your paper, use the abbreviation "Fig." even at the beginning of a sentence. Figures should be numbered with Arabic Numerals. Do not abbreviate "Table." Tables should be numbered with Roman Numerals.

L. SUBMITTING YOUR GRAPHICS

Figures should be submitted individually, separate from the manuscript in one of the file formats listed above in Section IV-C. Place figure captions below the figures; place table titles above the tables. Please do not include captions as part of the figures, or put them in "text boxes" linked to the figures. Also, do not place borders around the outside of your figures.

M. COLOR PROCESSING/PRINTING IN IEEE JOURNALS

All IEEE Transactions, Journals, and Letters allow an author to publish color figures on IEEE Xplore® at no charge, and automatically convert them to grayscale for print versions. In most journals, figures and tables may alternatively be printed in color if an author chooses to do so. Please note that this service comes at an extra expense to the author. If you intend to have print color graphics, include a note with your final paper indicating which figures or tables you would like to be handled that way, and stating that you are willing to pay the additional fee.

IX. CONCLUSION

Project Garuda demonstrates that a self-hosted, privacy-first, cloud-independent AI home security system can be built from consumer hardware for under US\$500 without compromising on real-time performance or modern security primitives. The

core engineering claim of the system is that a Raspberry Pi 5 paired with a Hailo-8L AI HAT+ can host the full inference stack – a custom-fine-tuned YOLOv8s detector for four threat-relevant classes, a MobileNetV2 classifier for *Safe/Weapon* verification, and a MiDaS Small depth estimator for monocular anti-spoof – at 52.22 FPS and 18.38 ms NPU latency, well inside the real-time envelope for continuous surveillance, while leaving the CPU free to run the FastAPI web server, the Narada voice assistant, the presence monitor, the deadman watchdog, and the logging engine concurrently.

On the model side, the progression from a 359-image v1 ($mAP_{50} = 0.465$) to the 4,982-image v5 ($mAP_{50} = 0.847$) with a +0.030 positive validation-to-test generalisation gap confirms that dataset scale and diversity – not architectural changes – are the dominant lever for robust real-world performance, and that the COCO baseline of 0.0002 mAP on this 4-class task is essentially random, reinforcing the need for targeted fine-tuning on security-relevant classes. The low off-diagonal values in the confusion matrix (≤ 0.03) and the alignment of the peak F1 confidence with YOLO's default 0.25 threshold mean the model ships without requiring any post-deployment threshold tuning.

On the systems side, the multi-engine architecture – alert fan-out with 60-second SMTP cooldown, WebSocket push, AES-256-CBC clip encryption, and SSH exfiltration; PBKDF2-SHA256 with 600,000 iterations and OTP 2FA; the 180-second deadman watchdog; six lock-guarded and scheduler-driven operating modes; ARP-based LAN presence; and the SQLite-backed offline event sync – closes the gap identified in the introduction between DIY motion-only setups and cloud-dependent professional systems. Every event reaches the dashboard in real time over WebSocket or WebRTC, and every event is also persisted to the multi-tier logger for offline catch-up, giving browser clients a coherent view of the system state even after arbitrary network outages.

The broader implication is that edge AI is no longer a compromise. With the right accelerator, the right post-training quantisation flow, and a disciplined software architecture, a device the size of a paperback can deliver detection quality, response latency, and security properties that previously required a cloud subscription and a server rack. Future extensions worth pursuing include expanding the class vocabulary beyond the four threat-relevant labels, exploring temporal models for activity recognition on top of the detector, integrating on-device TTS so that Narada remains functional when the public network is unavailable, and replacing the current static MiDaS variance threshold with a learned liveness head that is co-trained with the detector.

If you have multiple appendices, use the `\appendices` command below. If you have only one appendix, use `\appendix[Appendix Title]`

APPENDIX A FOOTNOTES

Number footnotes separately in superscript numbers.¹ Place the actual footnote at the bottom of the column in which it is cited; do not put footnotes in the reference list (endnotes). Use letters for table footnotes (see Table 11).

APPENDIX B SUBMITTING YOUR PAPER FOR REVIEW

A. FINAL STAGE

When your article is accepted, you can submit the final files, including figures, tables, and photos, per the journal's guidelines through the submission system used to submit the article. You may use *Zip* for large files, or compress files using *Compress*, *Pkzip*, *Stuffit*, or *Gzip*.

In addition, designate one author as the "corresponding author." This is the author to whom proofs of the paper will be sent. Proofs are sent to the corresponding author only.

B. REVIEW STAGE USING IEEE AUTHOR PORTAL

Article contributions to IEEE Access should be submitted electronically on the IEEE Author Portal. For more information, please visit <https://ieeaccess.ieee.org/>.

Along with other information, you will be asked to select the subject from a pull-down list. There are various steps to the submission process; you must complete all steps for a complete submission. At the end of each step you must click "Save and Continue"; just uploading the paper is not sufficient. After the last step, you should see a confirmation that the submission is complete. You should also receive an e-mail confirmation. For inquiries regarding the submission of your article, please contact ieeaccess@ieee.org.

The manuscript should be prepared in a double column, single-spaced format using a required IEEE Access template. A Word or LaTeX file and a PDF file are both required upon submission in the IEEE Author Portal.

C. FINAL STAGE USING IEEE AUTHOR PORTAL

Upon acceptance, you will receive an email with specific instructions

Designate the author who submitted the manuscript on IEEE Author Portal as the "corresponding author." This is the only author to whom proofs of the paper will be sent.

D. COPYRIGHT FORM

Authors must submit an electronic IEEE Copyright Form (eCF) upon submitting their final manuscript files. You can access the eCF system through your manuscript submission system or through the Author Gateway. You are responsible for obtaining any necessary approvals and/or security clearances. For additional information on intellectual property rights, visit the

¹It is recommended that footnotes be avoided (except for the unnumbered footnote with the receipt date on the first page). Instead, try to integrate the footnote information into the text.

IEEE Intellectual Property Rights department web page at http://www.ieee.org/publications_standards/publications/rights/index.html.

APPENDIX C IEEE PUBLISHING POLICY

The general IEEE policy requires that authors should only submit original work that has neither appeared elsewhere for publication, nor is under review for another refereed publication. The submitting author must disclose all prior publication(s) and current submissions when submitting a manuscript. Do not publish "preliminary" data or results. To avoid any delays in publication, please be sure to follow these instructions. Final submissions should include source files of your accepted manuscript, high quality graphic files, and a formatted pdf file. If you have any questions regarding the final submission process, please contact the administrative contact for the journal. author is responsible for obtaining agreement of all coauthors and any consent required from employers or sponsors before submitting an article.

The IEEE Access Editorial Office does not publish conference records or proceedings, but can publish articles related to conferences that have undergone rigorous peer review. Minimally, two reviews are required for every article submitted for peer review.

APPENDIX D PUBLICATION PRINCIPLES

Authors should consider the following points:

- 1) Technical papers submitted for publication must advance the state of knowledge and must cite relevant prior work.
- 2) The length of a submitted paper should be commensurate with the importance, or appropriate to the complexity, of the work. For example, an obvious extension of previously published work might not be appropriate for publication or might be adequately treated in just a few pages.
- 3) Authors must convince both peer reviewers and the editors of the scientific and technical merit of a paper; the standards of proof are higher when extraordinary or unexpected results are reported.
- 4) Because replication is required for scientific progress, papers submitted for publication must provide sufficient information to allow readers to perform similar experiments or calculations and use the reported results. Although not everything need be disclosed, a paper must contain new, useable, and fully described information. For example, a specimen's chemical composition need not be reported if the main purpose of a paper is to introduce a new measurement technique. Authors should expect to be challenged by reviewers if the results are not supported by adequate data and critical details.
- 5) Papers that describe ongoing work or announce the latest technical achievement, which are suitable for

presentation at a professional conference, may not be appropriate for publication.

APPENDIX E REFERENCE EXAMPLES

- *Basic format for books:*
J. K. Author, "Title of chapter in the book," in *Title of His Published Book*, xth ed. City of Publisher, (only U.S. State), Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx–xxx.
See [1], [2].
- *Basic format for periodicals:*
J. K. Author, "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, pp. xxx–xxx, Abbrev. Month, year, DOI: 10.1109.XXX.123456.
See [3]– [5].
- *Basic format for reports:*
J. K. Author, "Title of report," Abbrev. Name of Co., City of Co., Abbrev. State, Country, Rep. xxx, year.
See [6], [7].
- *Basic format for handbooks:*
Name of Manual/Handbook, x ed., Abbrev. Name of Co., City of Co., Abbrev. State, Country, year, pp. xxx–xxx.
See [8], [9].
- *Basic format for books (when available online):*
J. K. Author, "Title of chapter in the book," in *Title of Published Book*, xth ed. City of Publisher, State, Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx–xxx. [Online]. Available: <http://www.web.com>
See [10]– [13].
- *Basic format for journals (when available online):*
J. K. Author, "Name of paper," *Abbrev. Title of Periodical*, vol. x, no. x, pp. xxx–xxx, Abbrev. Month, year, Accessed on: Month, Day, year, DOI: 10.1109.XXX.123456, [Online].
See [14]– [16].
- *Basic format for papers presented at conferences (when available online):*
J.K. Author. (year, month). Title. presented at abbrev. conference title. [Type of Medium]. Available: site/path/file
See [17].
- *Basic format for reports and handbooks (when available online):*
J. K. Author. "Title of report," Company. City, State, Country. Rep. no., (optional: vol./issue), Date. [Online] Available: site/path/file
See [18], [19].
- *Basic format for computer programs and electronic documents (when available online):*
Legislative body. Number of Congress, Session. (year, month day). *Number of bill or resolution*, Title. [Type of medium]. Available: site/path/file
See [20].
- *Basic format for patents (when available online):*
Name of the invention, by inventor's name. (year, month day). Patent Number [Type of medium]. Available: site/path/file
See [21].
- *Basic format for conference proceedings (published):*
J. K. Author, "Title of paper," in *Abbreviated Name of Conf.*, City of Conf., Abbrev. State (if given), Country, year, pp. xxxxx.
See [22].
- *Example for papers presented at conferences (unpublished):*
See [23].
- *Basic format for patents:*
J. K. Author, "Title of patent," U.S. Patent x xxx xxx, Abbrev. Month, day, year.
See [24].
- *Basic format for theses (M.S.) and dissertations (Ph.D.):*
 - 1) J. K. Author, "Title of thesis," M.S. thesis, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.
 - 2) J. K. Author, "Title of dissertation," Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.
- See [25], [26].
- *Basic format for the most common types of unpublished references:*
 - 1) J. K. Author, private communication, Abbrev. Month, year.
 - 2) J. K. Author, "Title of paper," unpublished.
 - 3) J. K. Author, "Title of paper," to be published.
- See [27]– [29].
- *Basic formats for standards:*
 - 1) *Title of Standard*, Standard number, date.
 - 2) *Title of Standard*, Standard number, Corporate author, location, date.
- See [30], [31].
- *Article number in reference examples:*
See [32], [33].
- *Example when using et al.:*
See [34].

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in American English is without an "e" after the "g." Use the singular heading even if you have many acknowledgments. Avoid expressions such as "One of us (S.B.A.) would like to thank" Instead, write "F. A. Author thanks" In most cases, sponsor and financial support acknowledgments are placed in the unnumbered footnote on the first page, not here.

REFERENCES

- [1] G. O. Young, "Synthetic structure of industrial plastics," in *Plastics*, 2nd ed., vol. 3, J. Peters, Ed. New York, NY, USA: McGraw-Hill, 1964, pp. 15–64.
- [2] W.-K. Chen, *Linear Networks and Systems*. Belmont, CA, USA: Wadsworth, 1993, pp. 123–135.

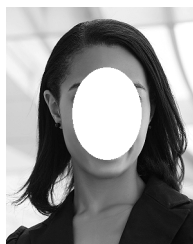
- [3] J. U. Duncombe, "Infrared navigation—Part I: An assessment of feasibility," *IEEE Trans. Electron Devices*, vol. ED-11, no. 1, pp. 34–39, Jan. 1959, 10.1109/TED.1961.2628402.
- [4] E. P. Wigner, "Theory of traveling-wave optical laser," *Phys. Rev.*, vol. 134, pp. A635–A646, Dec. 1965.
- [5] E. H. Miller, "A note on reflector arrays," *IEEE Trans. Antennas Propagat.*, to be published.
- [6] E. E. Reber, R. L. Michell, and C. J. Carter, "Oxygen absorption in the earth's atmosphere," Aerospace Corp., Los Angeles, CA, USA, Tech. Rep. TR-0200 (4230-46)-3, Nov. 1988.
- [7] J. H. Davis and J. R. Cogdell, "Calibration program for the 16-foot antenna," Elect. Eng. Res. Lab., Univ. Texas, Austin, TX, USA, Tech. Memo. NGL-006-69-3, Nov. 15, 1987.
- [8] *Transmission Systems for Communications*, 3rd ed., Western Electric Co., Winston-Salem, NC, USA, 1985, pp. 44–60.
- [9] *Motorola Semiconductor Data Manual*, Motorola Semiconductor Products Inc., Phoenix, AZ, USA, 1989.
- [10] G. O. Young, "Synthetic structure of industrial plastics," in *Plastics*, vol. 3, Polymers of Hexadromicon, J. Peters, Ed., 2nd ed. New York, NY, USA: McGraw-Hill, 1964, pp. 15–64. [Online]. Available: <http://www.bookref.com>.
- [11] *The Founders' Constitution*, Philip B. Kurland and Ralph Lerner, eds., Chicago, IL, USA: Univ. Chicago Press, 1987. [Online]. Available: <http://press-pubs.uchicago.edu/founders/>
- [12] The Terahertz Wave eBook. ZOmega Terahertz Corp., 2014. [Online]. Available: http://dl.z-thz.com/eBook/zomegabookpdf_1206_sr.pdf. Accessed on: May 19, 2014.
- [13] Philip B. Kurland and Ralph Lerner, eds., *The Founders' Constitution*. Chicago, IL, USA: Univ. of Chicago Press, 1987, Accessed on: Feb. 28, 2010, [Online]. Available: <http://press-pubs.uchicago.edu/founders/>
- [14] J. S. Turner, "New directions in communications," *IEEE J. Sel. Areas Commun.*, vol. 13, no. 1, pp. 11–23, Jan. 1995.
- [15] W. P. Risk, G. S. Kino, and H. J. Shaw, "Fiber-optic frequency shifter using a surface acoustic wave incident at an oblique angle," *Opt. Lett.*, vol. 11, no. 2, pp. 115–117, Feb. 1986.
- [16] P. Kopyt et al., "Electric properties of graphene-based conductive layers from DC up to terahertz range," *IEEE THz Sci. Technol.*, to be published. DOI: 10.1109/TTHZ.2016.2544142.
- [17] PROCESS Corporation, Boston, MA, USA. Intranets: Internet technologies deployed behind the firewall for corporate productivity. Presented at INET96 Annual Meeting. [Online]. Available: <http://home.process.com/Intranets/wp2.htm>
- [18] R. J. Hijmans and J. van Etten, "Raster: Geographic analysis and modeling with raster data," R Package Version 2.0-12, Jan. 12, 2012. [Online]. Available: <http://CRAN.R-project.org/package=raster>
- [19] Teralyzer. Lytera UG, Kirchhain, Germany [Online]. Available: http://www.lytera.de/Terahertz_THz_Spectroscopy.php?id=home, Accessed on: Jun. 5, 2014.
- [20] U.S. House. 102nd Congress, 1st Session. (1991, Jan. 11). *H. Con. Res. 1, Sense of the Congress on Approval of Military Action*. [Online]. Available: LEXIS Library: GENFED File: BILLS
- [21] Musical toothbrush with mirror, by L.M.R. Brooks. (1992, May 19). Patent D 326 189 [Online]. Available: NEXIS Library: LEXPAT File: DES
- [22] D. B. Payne and J. R. Stern, "Wavelength-switched passively coupled single-mode optical network," in *Proc. IOOC-ECOC*, Boston, MA, USA, 1985, pp. 585–590.
- [23] D. Ebehard and E. Voges, "Digital single sideband detection for interferometric sensors," presented at the 2nd Int. Conf. Optical Fiber Sensors, Stuttgart, Germany, Jan. 2-5, 1984.
- [24] G. Brandli and M. Dick, "Alternating current fed power supply," U.S. Patent 4 084 217, Nov. 4, 1978.
- [25] J. O. Williams, "Narrow-band analyzer," Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, USA, 1993.
- [26] N. Kawasaki, "Parametric study of thermal and chemical nonequilibrium nozzle flow," M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.
- [27] A. Harrison, private communication, May 1995.
- [28] B. Smith, "An approach to graphs of linear forms," unpublished.
- [29] A. Brahms, "Representation error for real numbers in binary computer arithmetic," IEEE Computer Group Repository, Paper R-67-85.
- [30] IEEE Criteria for Class IE Electric Systems, IEEE Standard 308, 1969.
- [31] Letter Symbols for Quantities, ANSI Standard Y10.5-1968.
- [32] R. Fardel, M. Nagel, F. Nuesch, T. Lippert, and A. Wokaun, "Fabrication of organic light emitting diode pixels by laser-assisted forward transfer," *Appl. Phys. Lett.*, vol. 91, no. 6, Aug. 2007, Art. no. 061103.
- [33] J. Zhang and N. Tansu, "Optical gain and laser characteristics of InGaN quantum wells on ternary InGaN substrates," *IEEE Photon. J.*, vol. 5, no. 2, Apr. 2013, Art. no. 2600111
- [34] S. Azodolmolky et al., Experimental demonstration of an impairment aware network planning and operation tool for transparent/translucent optical networks," *J. Lightw. Technol.*, vol. 29, no. 4, pp. 439–448, Sep. 2011.



FIRST A. AUTHOR received the B.S. and M.S. degrees in aerospace engineering from the University of Virginia, Charlottesville, in 2001 and the Ph.D. degree in mechanical engineering from Drexel University, Philadelphia, PA, in 2008.

From 2001 to 2004, he was a Research Assistant with the Princeton Plasma Physics Laboratory. Since 2009, he has been an Assistant Professor with the Mechanical Engineering Department, Texas A&M University, College Station. He is the author of three books, more than 150 articles, and more than 70 inventions. His research interests include high-pressure and high-density nonthermal plasma discharge processes and applications, microscale plasma discharges, discharges in liquids, spectroscopic diagnostics, plasma propulsion, and innovation plasma applications. He is an Associate Editor of the journal *Earth, Moon, Planets*, and holds two patents.

Dr. Author was a recipient of the International Association of Geomagnetism and Aeronomy Young Scientist Award for Excellence in 2008, and the IEEE Electromagnetic Compatibility Society Best Symposium Paper Award in 2011.



SECOND B. AUTHOR (M'76–SM'81–F'87) and all authors may include biographies. Biographies are often not included in conference-related papers. This author became a Member (M) of IEEE in 1976, a Senior Member (SM) in 1981, and a Fellow (F) in 1987. The first paragraph may contain a place and/or date of birth (list place, then date). Next, the author's educational background is listed. The degrees should be listed with type of degree in what field, which institution, city, state, and country, and year the degree was earned. The author's major field of study should be lower-cased.

The second paragraph uses the pronoun of the person (he or she) and not the author's last name. It lists military and work experience, including summer and fellowship jobs. Job titles are capitalized. The current job must have a location; previous positions may be listed without one. Information concerning previous publications may be included. Try not to list more than three books or published articles. The format for listing publishers of a book within the biography is: title of book (publisher name, year) similar to a reference. Current and previous research interests end the paragraph.

The third paragraph begins with the author's title and last name (e.g., Dr. Smith, Prof. Jones, Mr. Kajor, Ms. Hunter). List any memberships in professional societies other than the IEEE. Finally, list any awards and work for IEEE committees and publications. If a photograph is provided, it should be of good quality, and professional-looking. Following are two examples of an author's biography.

THIRD C. AUTHOR, JR. (M'87) received the B.S. degree in mechanical engineering from National Chung Cheng University, Chiayi, Taiwan, in 2004 and the M.S. degree in mechanical engineering from National Tsing Hua University, Hsinchu, Taiwan, in 2006. He is currently pursuing the Ph.D. degree in mechanical engineering at Texas A&M University, College Station, TX, USA.

From 2008 to 2009, he was a Research Assistant with the Institute of Physics, Academia Sinica, Tapei, Taiwan. His research interest includes the development of surface processing and biological/medical treatment techniques using nonthermal atmospheric pressure plasmas, fundamental study of plasma sources, and fabrication of micro- or nanostructured surfaces.

Mr. Author's awards and honors include the Frew Fellowship (Australian Academy of Science), the I. I. Rabi Prize (APS), the European Frequency and Time Forum Award, the Carl Zeiss Research Award, the William F. Meggers Award and the Adolph Lomb Medal (OSA).

• • •